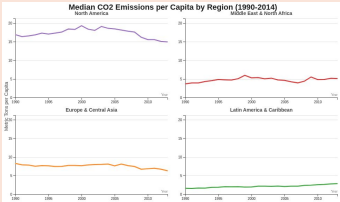
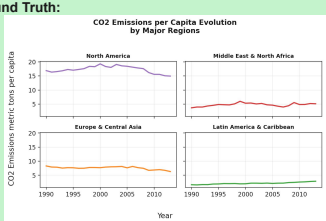
DV-Evol Good Case d3.js

Question: Convert the 1990–2014 regional CO2 emissions data from mean to country-level median aggregation and replace the stacked area chart with a faceted line chart using specific HEX color mappings.

Gemini-3-pro-preview:



Ground Truth:



Plot code:

```
const colors = {"North America":"#9467bd","Middle East & North Africa":"#d62728","Europe & Central Asia":"#ff7f0e","Latin America & Caribbean":"#2ca02c"};
const svg = d3.select("body").append("svg").attr("width", 960).attr("height", 600);
const x = d3.scaleLinear().domain([1990, 2014]).range([0, 380]);

d3.csv("result.csv").then(data => {
  const y = d3.scaleLinear().domain([0, d3.max(data, d => +d.Val)]).range([200, 0]);
  d3.groups(data, d => d.Region).forEach(([reg, v], i) => {
    const g = svg.append("g").attr("transform", `translate${(i%2)*480+50},${Math.floor(i/2)*280+50}` );
    g.append("g").attr("transform", "translate(0,200)").call(d3.axisBottom(x).ticks(5).tickFormat(d3.format("d")));
    g.append("g").call(d3.axisLeft(y).ticks(5));
    g.append("path").datum(v).attr("fill", "none").attr("stroke", colors[reg]).attr("stroke-width", 3)
      .attr("d", d3.line().x(d => x(+d.Year)).y(d => y(+d.Val)));
    g.append("text").text(reg).attr("x", 190).attr("y", -10).attr("text-anchor", "middle").style("font-weight", "bold");
  });
});
```

Error Analysis: 1. Data Integrity

✅1.1 Trend/Pattern: 1/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

✅1.2 Completeness: 1/1 Critical data series or specific years are missing from the visualization.

✅1.3 Label Accuracy: 1/1 Axis tick values and labels do not precisely match the original data points.

✅1.4 Data Consistency: 1/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

✅2.5 Element Spacing: 1/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

✅3.1 Labels: 1/1 Legend and title positioning are misaligned compared to the standard layout.

✅3.2 Readability: 1/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

✅3.3 Axis Alignment and Spacing: 1/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

✅3.4 Element Alignment: 1/1 Graphical components fail to align precisely with their respective axis ticks.

✅3.5 Proportionality and Aspect Ratio: 1/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

✅3.6 Overall Balance: 1/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

Score: 13/16